文献汇报

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Smoke Detection on Video Sequences Using 3D Convolutional Neural Networks

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方法概述

开发了一个基于快速RCNN和3D CNN的联合检测框架。提出了一种改进的非最大聚合的快速RCNN算法，实现了基于静态空间信息的烟雾目标定位。然后，3D CNN结合动态时空信息实现烟雾识别。

与常用的利用图像进行烟雾检测的CNN方法相比，3D CNN显著提高了识别精度。比较了三维CNN的不同网络结构和数据处理方法，包括慢融合和光流。在包含来自多个来源的烟雾视频的数据集上进行了测试，结果表明所提出的框架在烟雾定位和识别方面表现得非常好。

Faster-RCNN

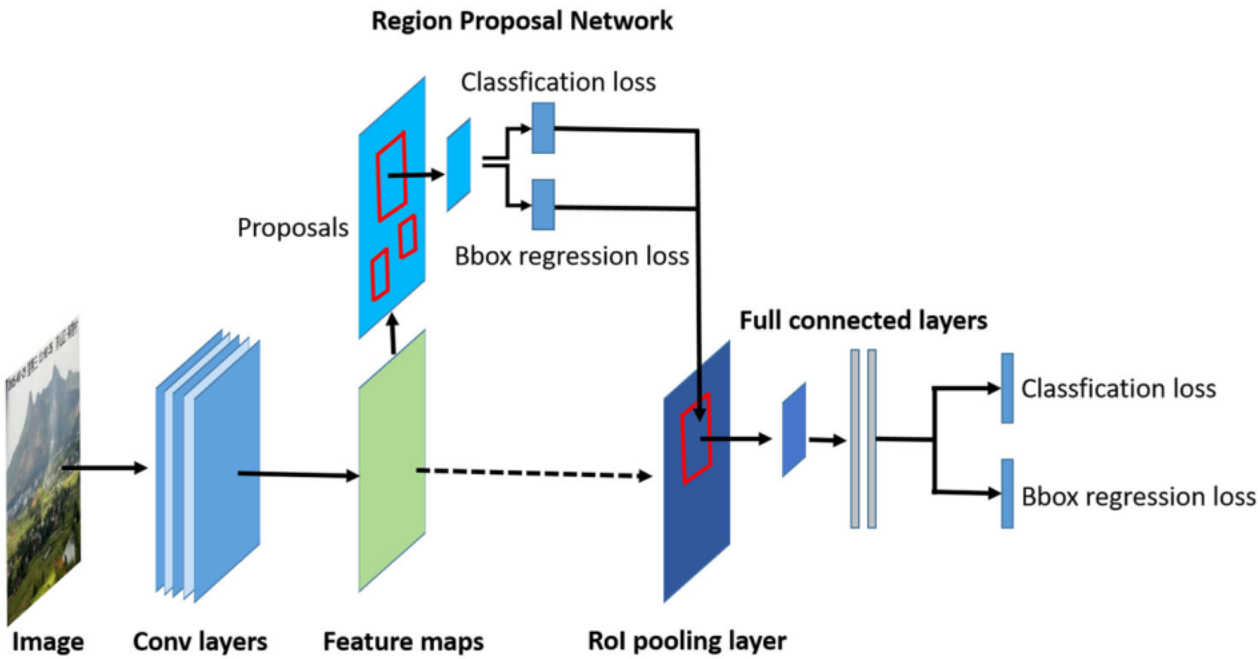


Figure 2. Structure of faster RCNN. Region proposal network proposes propose candidate regions.

3D 卷积

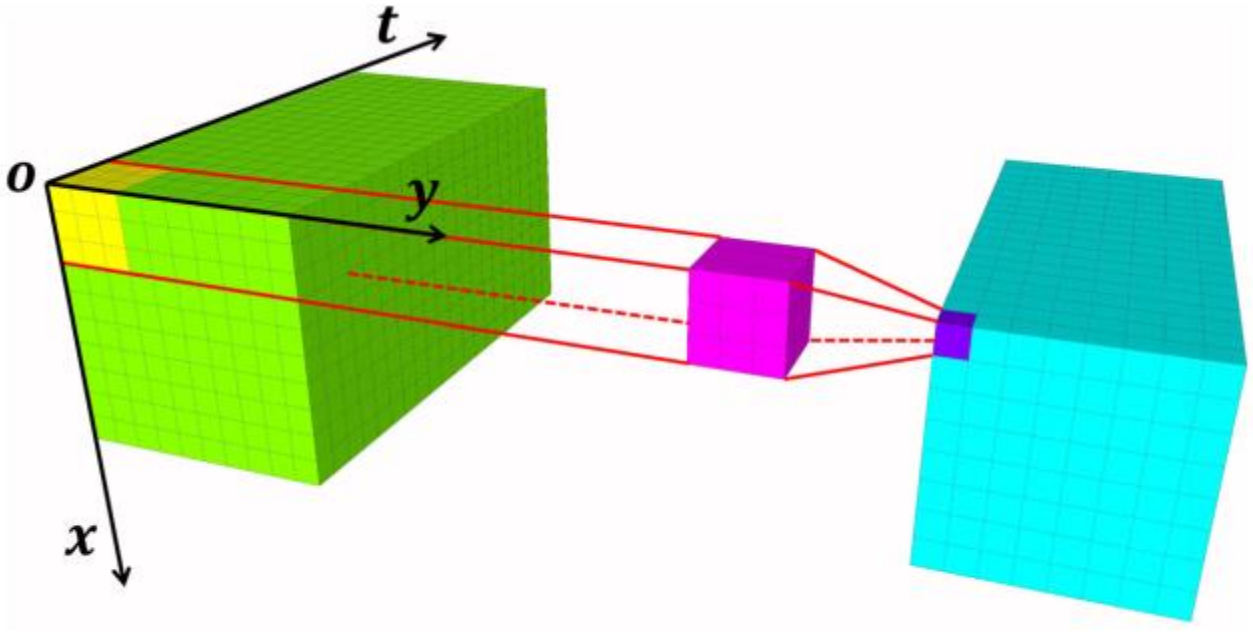


Figure 3. 3D convolution operation in video sequences.

网络框图

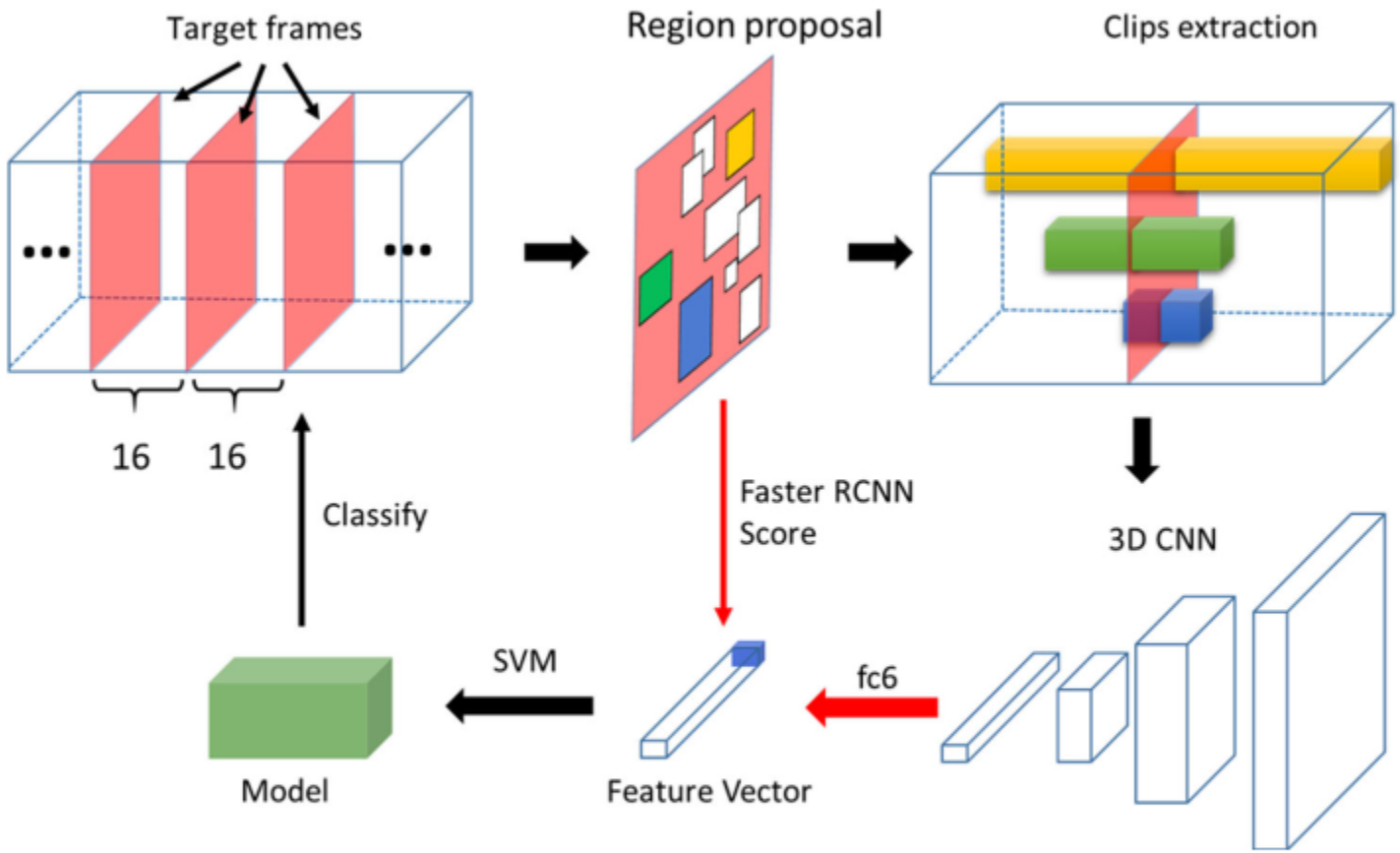
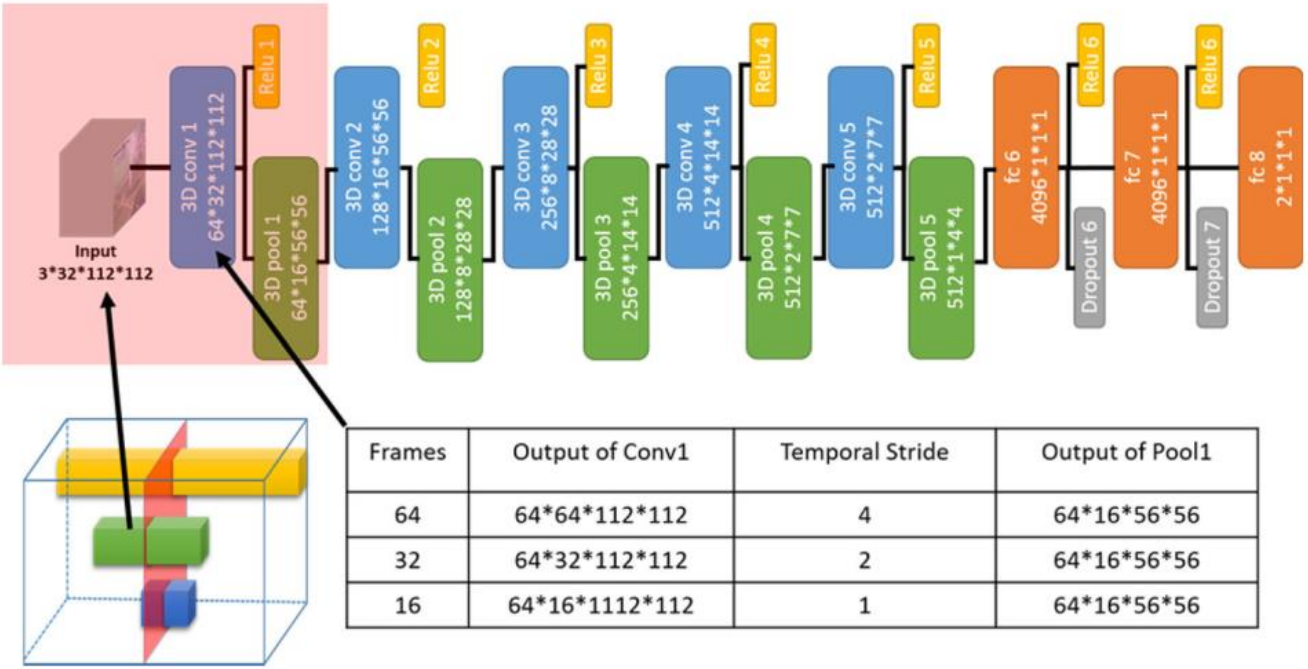


Figure 4. Overview of our joint detection framework for video smoke detection. Using faster RCNN to initially smoke detection on target frame.

网络框图



Frames	Output of Conv1	Temporal Stride	Output of Pool1
64	64*64*112*112	4	64*16*56*56
32	64*32*112*112	2	64*16*56*56
16	64*16*1112*112	1	64*16*56*56

Figure 8. Structure of 3D convolutional neural network. The input can be selected in three clip lengths.

在3D卷积使用中，一般输入是一个简短的16帧视频剪辑。然而，烟雾的扩散非常缓慢，特别是在森林等远距离监测环境中。在不到一秒的时间框架内，几乎看不到这种运动。除了16帧的剪辑外，我们还使用32帧和64帧的剪辑作为三维CNN的输入。所有剪辑帧的大小被调整到128*171。在训练中，输入维度是3*长度*128*171，随机大小为3*长度*112*112。

NMA

Table 1
NMS and Non-maximum Annexation

NMS	Rank 300 boxes by score
	If IoU(boundingbox$_i$, box$_j$) > 0.3, delete box$_j$
	Else if IoU(boundingbox$_i$, box$_j$) < 0.3, save box$_j$ as *boundingbox*$_{i+1}$
	If score of boundingbox$_i$ > 0.8 alarm
	Rank boxes by score
NMA	If score of box$_j$ < 0.01, delete box$_j$
	Else if $\sum_{i=1}^{I}$ IoU(boundingbox$_i$, box$_j$) == 0, save box$_j$ as *boundingbox*$_{i+1}$
	Else if IoU(boundingbox$_i$, box$_j$) < 0.6 &&
	$\sum_{k \neq i}^{I}$ IoU(boundingbox$_k$, boundingbox$_i$U box$_j$) == 0, save boundingbox$_i$U box$_j$ as boundingbox$_i$
	Take boundingbox$_i$($i = 1, 2 \ldots I$) as suspected smoke boxes

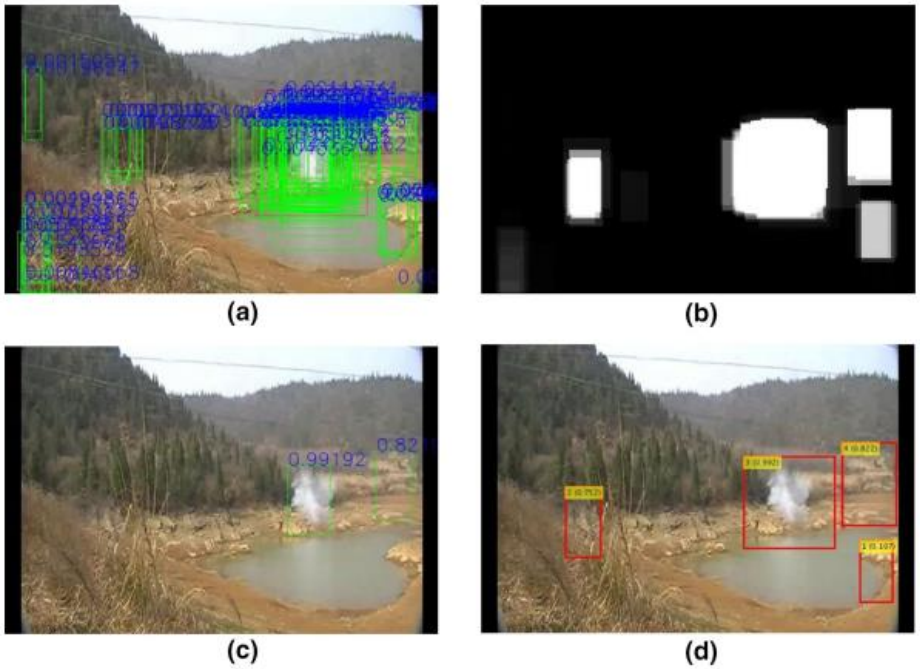


Figure 6. Difference between NMS and NMA. (a) 300 boxes proposed by RPN, (b) superposition of 300 boxes' scores, (c) bounding boxes generated by NMS, (d) bounding boxes generated by NMA.

数据增强

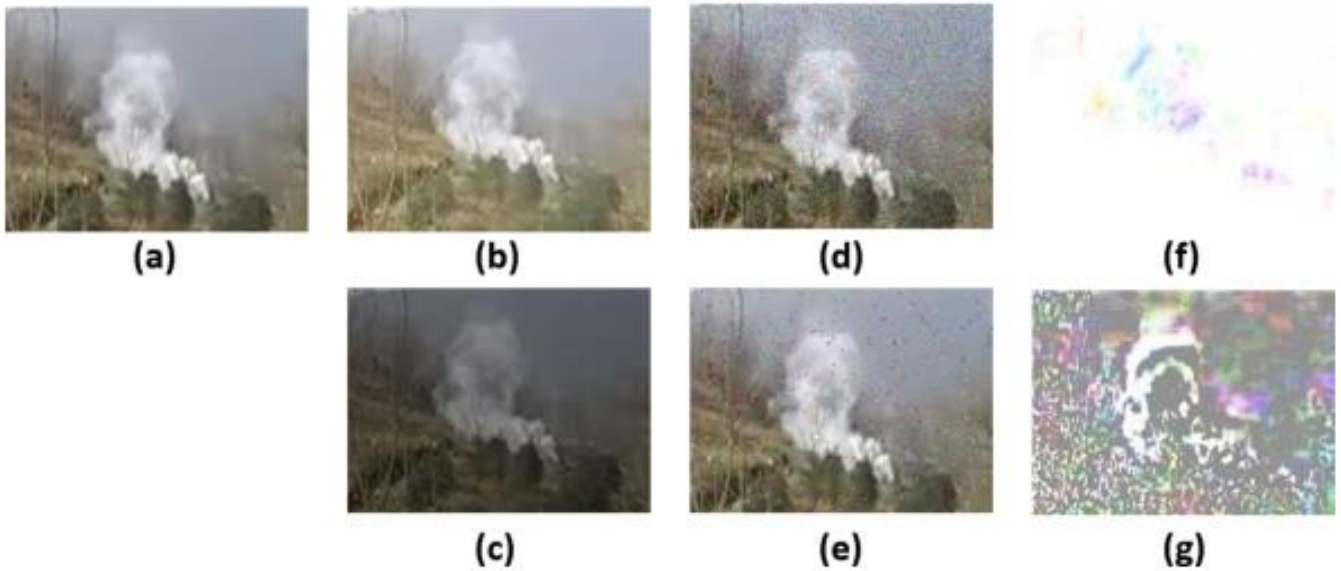


Figure 7. Example of data augmentation. (a) Original frame, (b) brightness increase, (c) brightness decrease, (d) Gaussian white noise, (e) salt-pepper noise, (f) optical flow frame, (g) background subtraction frame.

数据收集

Table 2
Statistics of the Experimental Dataset

Dataset	Videos	Durations (s)	Target Frames	Bounding boxes	Smoke boxes	Non-smoke boxes
Train	20	1471	2436	6568	2168	4400
Test smoke	18	1549	2402	4619	2858	1761
Test non-smoke	20	2393	3953	11,682	0	11,682

Table 3
Results of Faster RCNN and 3D CNN

Methods	AR (%)	DR (%)	PR (%)	FAR (%)
Faster RCNN threshold = 0.8				
Bounding box	75.71	**77.96**	40.09	24.76
Frame	59.34	**85.97**	47.89	56.84
Faster RCNN + C3D				
Bounding box	**93.74**	74.00	**88.42**	**2.06**
Frame	**91.88**	84.72	**93.18**	**3.77**

Bold values indicate the optimal value in the column, that is, the highest AR, DR, PR, or the lowest FAR

Table 4
Results of Training Model with Different Input Frames

Methods	AR (%)	DR (%)	PR (%)	FAR (%)
AlexNet	80.10	**89.05**	46.48	21.80
GoogleNet	85.72	86.14	56.03	14.37
3DCNN	**93.74**	74.00	**88.42**	**2.06**

Table 5
Results of Fine-Tuning Model with Different Input Frames

Methods	AR (%)	DR (%)	PR (%)	FAR (%)
Finetuning_16	92.89	**75.33**	82.59	3.38
Finetuning_32	92.49	64.77	89.51	1.61
Finetuning_64	**93.31**	67.63	**92.09**	**1.23**

Bold values indicate the optimal value in the column, that is, the highest AR, DR, PR, or the lowest FAR

Table 6
Results of Training Model with Different Input Frames

Methods	AR (%)	DR (%)	PR (%)	FAR (%)
training_16	**93.87**	**74.81**	88.46	2.08
training_32	93.74	74.00	88.42	2.06
training_64	93.56	72.36	**88.87**	**1.93**